

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester I Examination 2024-25 (NEP)

DEPARTMENT OF STATISTICS**Paper Code: STAMJ11 / STAMN11 — Descriptive Statistics**

Time: 2:30 Hours

Maximum Marks: 70

*Instructions: Answer any **Five** questions. Each question carrying equal marks (14 marks each). Question number 01 is **compulsory**. Each question should start from a fresh page.*

Question 1 (Compulsory)

- (a) What are the two senses in which the term Statistics is generally used? [3.5]
- (b) Define the terms: frequency, relative frequency, cumulative frequency (less than and more than type), and frequency density. [3.5]
- (c) The weights (in kg) of 8 persons are: 62.65, 64.92, 64.01, 121.22, 63.11, 69.01, 67.19, and 72.64. Find the average weight using a suitable form of average. Give reason for your choice. [3.5]
- (d) If the correlation coefficient between x and y is 0.5, what would be the correlation coefficient between $5x$ and $-3y$? [3.5]

Question 2

- (a) Define mean, median, and mode with their merits and demerits. Also discuss their specific use with suitable examples. Show that the sum of deviation taken from mean is always zero. [9]
- (b) The mean age of a group of 100 children is 9.35 years. The mean age of 25 of them is 8.75 years and that of another 65 is 10.51 years. What is the mean age of the remainder? [5]

Question 3

- (a) If x_1 and x_2 are two positive values of a variate, prove that their geometric mean is equal to the geometric mean of their arithmetic and harmonic means. If the arithmetic mean of two observations is 25 and their geometric mean is 15, find their harmonic mean and the two observations. [7]
- (b) You are given the following incomplete frequency distribution. It is known that the total frequency is 1000 and the median is 413.11. Find the missing frequencies and first quartile:

oprule Values	300–325	325–350	350–375	375–400	400–425	425–450	450–475	475–500
Frequency	5	17	80	?	326	?	88	9
ottomrule								

[7]

Question 4

- (a) Prove that the standard deviation calculated from two values x_1 and x_2 of a variable x is equal to half of their absolute difference. [4.5]
- (b) Find the standard deviation of the first n natural numbers. [4.5]
- (c) The mean and variance of n values $x_1, x_2, x_3, \dots, x_n$ are 0 and σ^2 respectively. If y is a variable defined as $y = x^2$, then show that $\text{avg } y = \sigma^2$. [5]

Question 5

- (a) Out of 400 observations, 100 observations have the value one and the rest of observations are zero. Find the standard deviation of 400 observations together. [3.5]
- (b) The mean of a certain distribution is 5. The second and third moment about the mean is 20 and 140 respectively. Find the third moment of the distribution about 10. [3.5]
- (c) If the first quartile is 142 and inter quartile range is 18, then find median assuming the distribution is symmetrical. [3.5]
- (d) Explain both the measure of Kurtosis and its different types. [3.5]

Question 6

- (a) Calculate the correlation coefficient and its probable error from the following results ($n = 40$):

$$\sum_{i=1}^{40} x_i = 280, \quad \sum_{i=1}^{40} y_i = 60, \quad \sum_{i=1}^{40} x_i^2 = 2384, \quad \sum_{i=1}^{40} y_i^2 = 117, \quad \sum_{i=1}^{40} x_i y_i = 438$$

[7]

- (b) What are regression lines? Explain why we have two regression lines and why these two lines are identical if correlation coefficient is $+1$ or -1 . Also show that the correlation coefficient is geometric mean of the regression coefficients. [7]

Question 7

- (a) In the following table, the recorded data shows the test scores made by 10 salesmen on an IQ test and their weekly sales:

Tests Scores on IQ	50	70	50	60	80	50	90	50	60	60
Weekly Sales (in '000 Rs.)	25	60	45	50	45	20	60	30	45	40

Calculate the Rank correlation coefficient between IQ and efficiency in sales. [5]

- (b) If A and B are independent attributes, $N = 24$, $(A) = 16$ and $(B) = 6$, find the ultimate class frequencies. [4.5]

- (c) In a college out of 200 students 150 are boys. In an examination 120 boys and 40 girls passed. Find Yule's coefficient of association between gender and success, also comment on the result obtained. [4.5]